

JOSH DELON

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Duren Sawit, Jakarta Timur, DKI Jakarta, Indonesia

Electrical Engineering graduate with hands- on experience in networking, cloud infrastructure, IoT, and automation using AWS, Terraform, and Linux.

Education

SMK Swasta YAPIM Balige - **Balige, North Sumatra, Indonesia** Jun 2019 - Jun 2022

High School Diploma in Computer and Network Engineering

- Studied fundamental network protocols including TCP/IP, DNS, and DHCP.
- Configured routers and switches using Cisco Packet Tracer.
- Participated in network installation during an internship program.

Institut Teknologi Del - **Laguboti, North Sumatra, Indonesia** Jul 2022 - Sep 2026

Bachelor of Electrical Engineering

- Member of Himpunan Mahasiswa Elektro (HME) and Del Robotics Club.
- Served as Leader of Majelis Permusyawaratan Anggota HME (MPA-HME), representing members in strategic decision-making.

Experience & Projects

Kominfo Toba - **Kominfo-Balige, Indonesia** Jul 2021 - Oct 2021

Internship - IT Support

During the internship program at the Toba Regency Communications and Informatics Office (Kominfo), I contributed to various activities supporting operations and public information management.

- Assisted in the management of documents and administration related to Kominfo's work programs.
- Participated in the maintenance of computer networks and hardware used by the office.
- Helped in the management of data and information in Kominfo's internal systems.
- Responsible for the repair and maintenance of videotron systems.

Inovasi Sistem Monitoring Konsumsi Energi Listrik pada Gedung Berbasis Internet of Things - **Del Institute of Technology – Laguboti, Indonesia** Nov 2024

Student

- Circuit Design: Connecting the SCT-013 current sensor and ZMPT101B voltage sensor to the ESP32 module for data collection.
- Application Development: Creating a user interface in the Blynk application to visually display energy consumption data.
- The IoT system successfully monitors energy consumption with 95% accuracy and provides real- time data through the Blynk application.

PT.Helber Elektronika Indonesia - **Daerah Istimewa Yogyakarta** Jun 2025 - Aug 2025

Internship

Electronic and Mechanical Production Staff

- Built a Python GUI control system for a 5-DOF robotic arm, integrating Arduino Nano and Bluetooth for real-time control and system validation in industrial automation learning prototypes.
- Supported Project Manager in QC and acceptance testing of Medical Compressors, conducting functional and pressure tests, defect analysis, and release documentation for regulated production decisions.
- Contributed to electronic & mechanical production and field installations, including BSC electrical setup, PCB soldering, sensor integration, and on-site system testing to ensure reliability and standards compliance.

Site-to-Site & Remote-Access VPN Integration using MikroTik - **Del Institute of Technology – Laguboti, Indonesia** Dec 2025

Student

- Design and implement a PPTP-based VPN architecture to integrate two campus laboratories on different subnets and provide secure remote access for external users.

Real-Time Solar Energy & Battery Health Monitoring for IoT Greenhouse - **Del Institute of Technology – Laguboti, Indonesia** Dec 2025

Student

- Designing and implementing an IoT-based energy monitoring system for off-grid greenhouses that monitors solar panel power output and battery health in real time. The system uses ESP32 and INA219 sensor nodes with MQTT communication to a local server and

Grafana visualization for energy performance analysis.

Implementation of a Real-Time pH and EC Monitoring System for IoT-Based Hydroponics with Data Visualization - **Del Institute of Technology – Laguboti, Indonesia** Oct 2025 - Jul 2026

Student

Final Project

- Designing a real-time hydroponic monitoring system based on Edge-IoT using an ESP32 as the sensor reader and a Raspberry Pi 4 as the local server.
- Building a data processing workflow using the MQTT protocol (Mosquitto), Telegraf, InfluxDB (a time-series database), and Grafana for an interactive visualization dashboard.
- Integrate Cloudflare Tunnel for secure remote access and the Telegram notification system for early warnings.

Independent Project - **Laguboti, Indonesia** May 2026

Cloud Infrastructure Developer

A self-directed project to build a private cloud storage infrastructure

- Deployed an Ubuntu 22.04 server on Amazon EC2 (t3.micro) using Terraform for automated Infrastructure as Code (IaC).
- Developed a web- based backend using Python Flask to facilitate cross- device file management operations, including uploading, downloading, and deleting files.
- Configured Amazon S3 as the primary object storage for scalable and reliable data retention, allowing seamless access from both mobile and desktop devices on the same network.
- Planned the integration of an authentication system to enhance data security and user access control.

Organizational Experience

Himpunan Mahasiswa Elektro(HME) - **Institut Teknologi Del, Laguboti, Indonesia** Nov 2022 - Sep 2026

Member

Electrical Engineering student association at Institut Teknologi Del.

- Contributed to educational development activities within the Electrical Engineering student association.

Wali Supervisor Program Cinta Almamater - **Institut Teknologi Del, Laguboti, Indonesia** Aug 2023

Supervisor

Program Cinta Almamater at Institut Teknologi Del.

- Supervised participants during the Program Cinta Almamater.

Del Robotics Club - **Institut Teknologi Del, Laguboti, Indonesia** Sep 2023 - Sep 2026

Member

Robotics club at Institut Teknologi Del.

- Participate in national or international robotics competitions to test the capabilities of the designed robots.
- Design and develop various types of robots, such as line follower robots, firefighting robots, or IoT-based robots.
- Become a Mechanic in the field of Indonesian Wheeled Robot Contest (KRSBIB)

Inauguration, Dies Natalis, and Graduation Committee 2023 - **Institut Teknologi Del, Laguboti, Indonesia** Sep 2023

General Section

Committee for the 2023 inauguration, Dies Natalis, and graduation events at Institut Teknologi Del.

Majelis Permusyawaratan Anggota HME - **Institut Teknologi Del, Laguboti, Indonesia** Oct 2024 - Oct 2025

Leader

Assembly of members within the Electrical Engineering student association at Institut Teknologi Del.

- Led strategic decision-making and represented members' voices in organizational meetings.

Skills and Other

- **Technical Skills**: PCB Design (Autodesk Eagle, Proteus), IoT Development (Arduino IDE, Python), Computer Networking (Cisco Packet Tracer, VirtualBox, Ubuntu, Kali Linux), Cloud Computing & DevOps (AWS EC2, S3, Lambda, RDS PostgreSQL, EKS, IAM, ECR), Containerization (Docker, Kubernetes, microk8s), Infrastructure as Code (Terraform, AWS CDK).
- **Programming Language**: C, Python, JavaScript, SQL.
- **Soft Skills**: Team Collaboration, Leadership, Problem-Solving, Time Management.